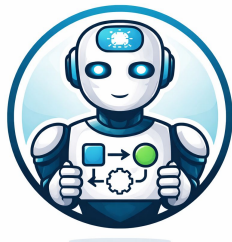


Agentic AI for Tech Professionals

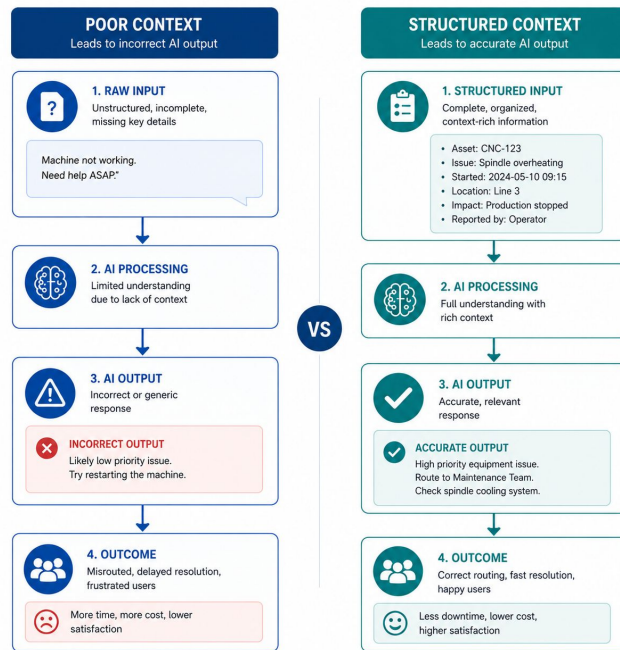
Context and RAG



Why Context Matters

AI quality depends on the context you provide

- AI does not “know” your company
- Context improves accuracy
- Poor context = hallucination
- Structured input improves consistency
- Enterprise AI needs controlled context



Why Context Matters (2)

LLMs are powerful,
but they do not automatically understand:

- your SOPs
- your production process
- your internal terminology

Why Context Matters (3)

Without context:

- ✗ generic answers
- ✗ hallucinations
- ✗ inconsistent responses

With proper context:

- ✓ accurate outputs
- ✓ enterprise relevance
- ✓ predictable behavior

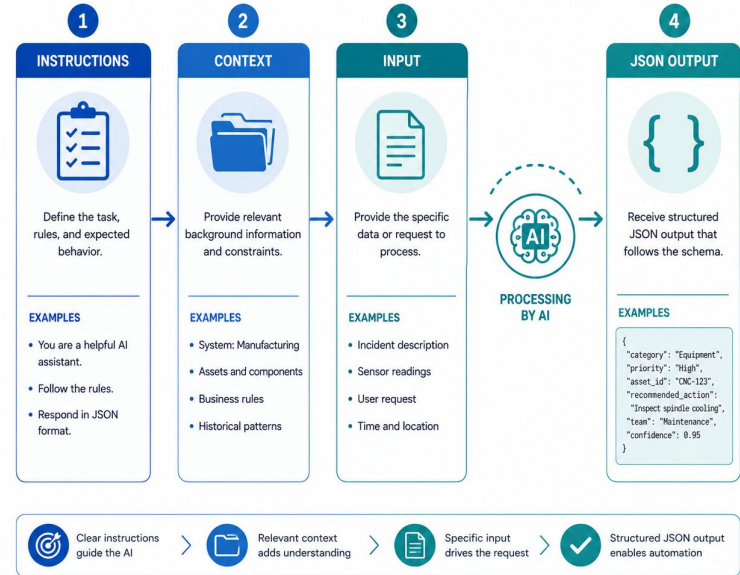
This is why context engineering is critical.

Structured Prompting

Well-structured prompts create stable AI systems

- Use clear instructions
- Define expected format
- Prefer JSON output
- Separate rules from input
- Reduce randomness

STRUCTURED AI PROMPT FLOW



Structured Prompting (2)

Good prompting is not:

👉 “Please help me”

Good prompting is:

- role
- task
- format
- rules
- input

Example:

Instead of:

✗ “Analyze this issue”

Use:

✓ “Classify into Equipment, IT, or Process and return JSON”

This transforms AI from:

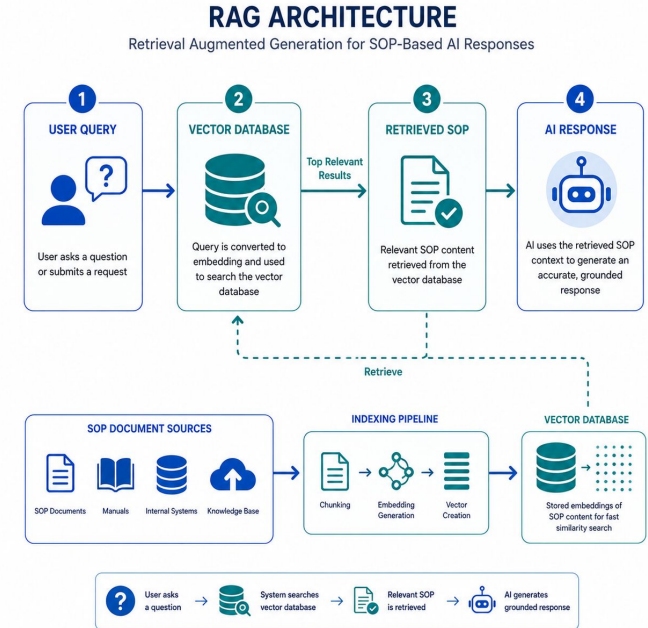
✗ chatbot

✓ reliable system component

Introduction to RAG

RAG allows AI to retrieve knowledge before responding

- RAG = Retrieval-Augmented Generation
- Retrieve relevant documents first
- AI answers using retrieved context
- Reduces hallucination
- Enables SOP-aware AI systems



Introduction to RAG (2)

RAG is one of the most important enterprise AI patterns.

Instead of relying only on AI memory:

👉 we retrieve real company knowledge first.

Example:

Issue:

“Cooling system overheating”

System retrieves:

👉 SOP document for cooling procedure

Then AI responds using:

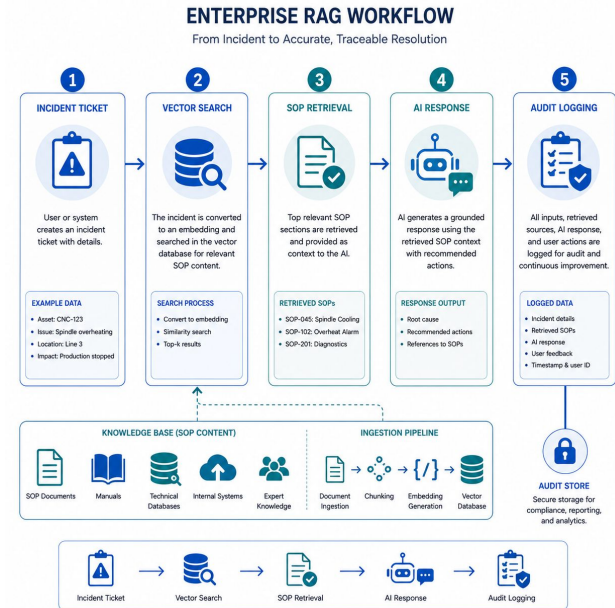
👉 actual company knowledge

This is how enterprise AI becomes trustworthy.

RAG Workflow in Manufacturing Support

AI retrieves SOPs before generating operational responses

- Input incident ticket
- Search knowledge base
- Retrieve relevant SOP
- Generate guided response
- Log for audit/compliance



RAG Workflow in Manufacturing Support (2)

This is your future enterprise system.

Input:

“Machine overheating”

Workflow:

1. Classify issue
2. Search SOP database
3. Retrieve maintenance procedure
4. Generate response
5. Store decision log

This becomes:

👉 intelligent operational support system

What is Memory in Agentic AI?

Memory allows AI systems to retain context, history, and operational knowledge

- AI needs context to make better decisions
- Memory stores previous information
- Enables workflow continuity
- Improves response relevance
- Critical for enterprise AI systems

What is Memory in Agentic AI? (2)

Without memory,
AI behaves like:

👉 a person with no recollection of the past.

Every request becomes:

✗ isolated

Memory allows AI systems to:

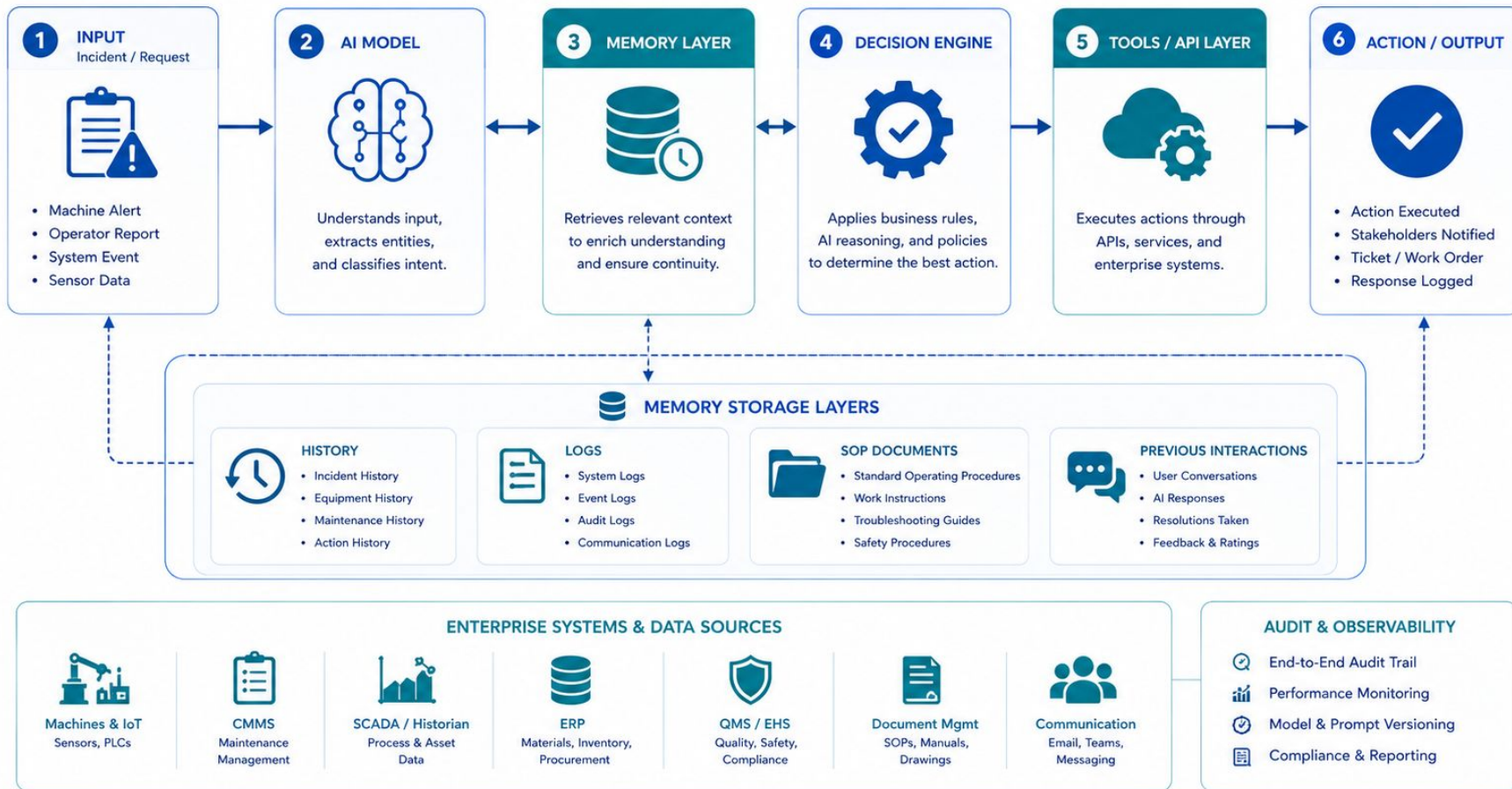
- remember previous interactions
- retrieve operational knowledge
- maintain workflow continuity
- improve decision quality

This becomes essential for:

👉 enterprise operational systems.

AI Workflow with Connected Memory

Intelligent workflows powered by context, history, and enterprise knowledge



Types of Memory in AI Systems

Different memory layers support different enterprise AI capabilities

- Short-term memory → current context
- Long-term memory → historical records
- Knowledge memory → SOP/RAG documents
- Workflow memory → execution state
- User memory → preferences/history

AI Memory Architecture (Layered)

Structured memory layers powering context, continuity, and intelligent decisions



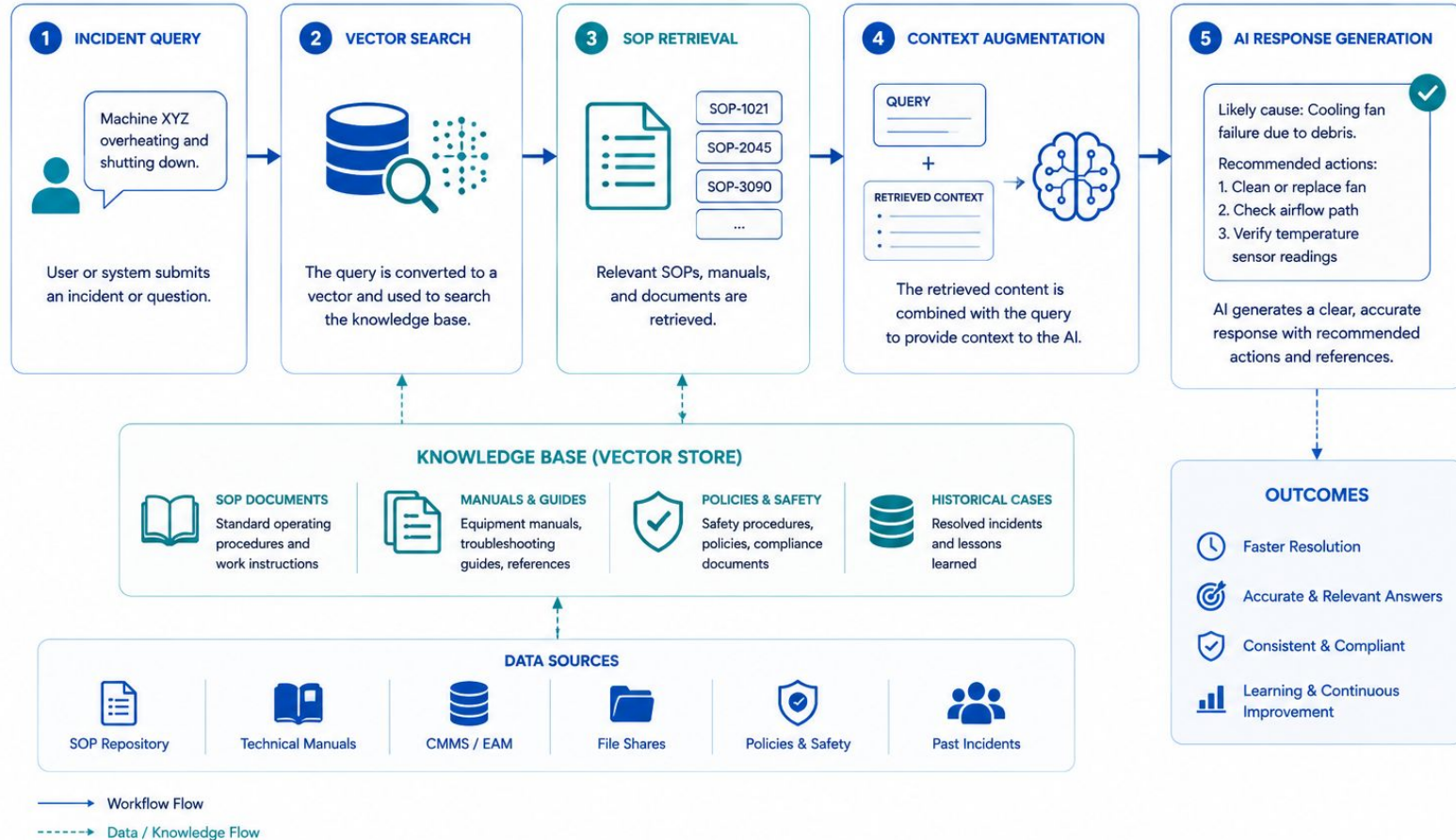
Memory + RAG (Retrieval-Augmented Generation)

RAG gives AI access to enterprise knowledge during decision-making

- Retrieves relevant knowledge dynamically
- Reduces hallucination risk
- Uses vector databases for search
- Injects context into prompts
- Enables SOP-aware AI systems

RAG Workflow for Incident Resolution

Retrieve. Augment. Generate. Resolve.



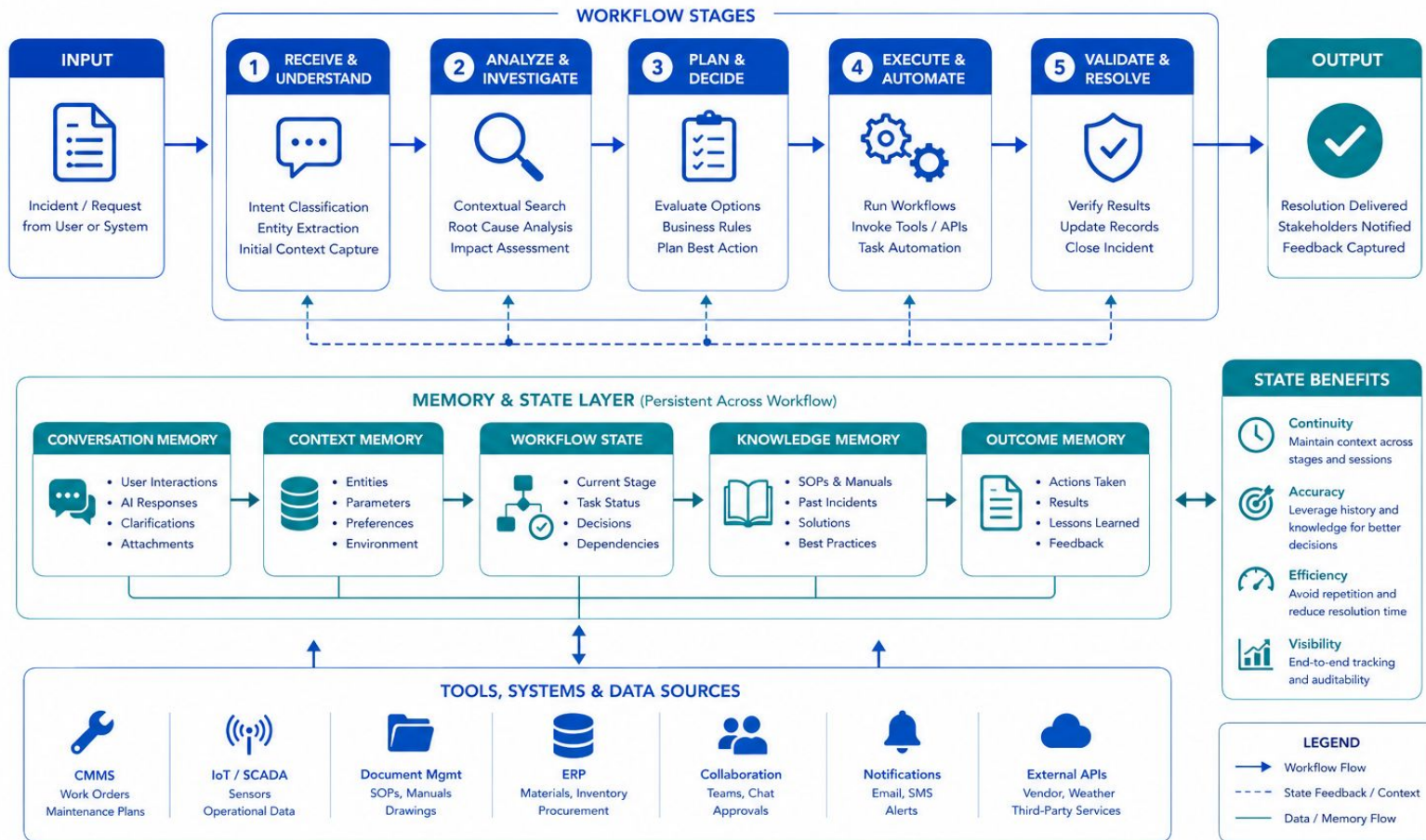
Workflow Memory in Enterprise Automation

Workflows need memory to track state and operational progress

- Stores workflow execution status
- Tracks approvals and escalations
- Enables retries and monitoring
- Supports long-running workflows
- Important for enterprise orchestration

Workflow Orchestration with Memory & State Tracking

Maintain context. Track state. Orchestrate outcomes.



Enterprise Manufacturing AI Architecture

Intelligent incident resolution with connected memory and insights

